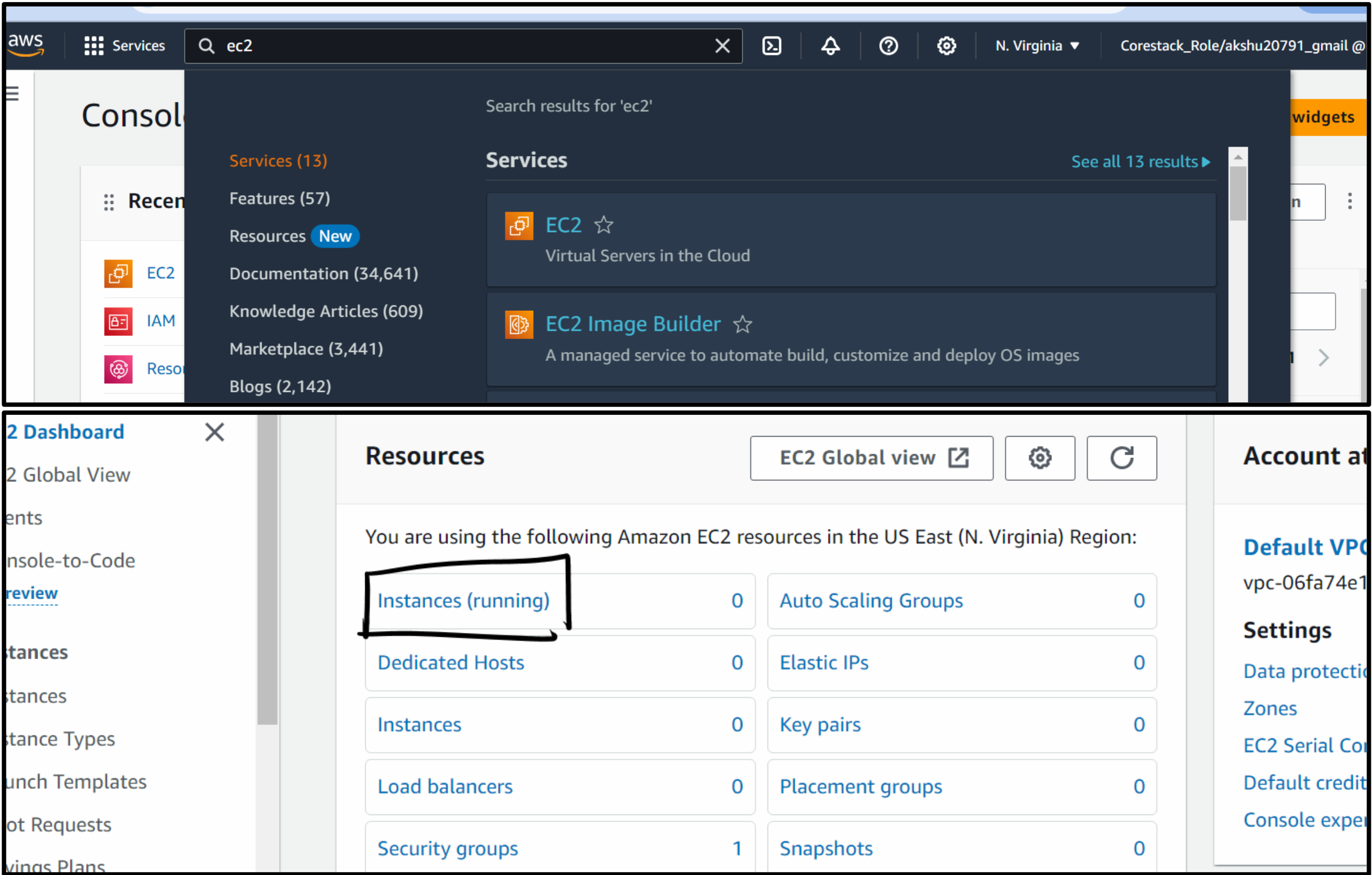


Lesson 01 Demo 01

INSTALLING AND CONFIGURING ANSIBLE



Click on Launch instances

Create key pair

Key pair name

Key pairs allow you to connect to your instance securely.

ansible-key

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

☒ RSA
RSA encrypted private and public key pair

☐ ED25519
ED25519 encrypted private and public key pair

Private key file format

☒ .pem
For use with OpenSSH

☐ .ppk
For use with PuTTY

Cancel

Create key pair

The key would be downloaded to the machine

In Firewall ,

Enable

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow sp your instance.

☒ Create security group ☐ Select existing security group

We'll create a new security group called '**launch-wizard-1**' with the following rules:

- ☒ Allow SSH traffic from

Helps you connect to your instance

Anywhere
0.0.0.0/0 ▼
- ☐ Allow HTTPS traffic from the internet

To set up an endpoint, for example when creating a web server
- ☒ Allow HTTP traffic from the internet

To set up an endpoint, for example when creating a web server

Change the number of instances to 3 and launch the instances. We will consider one machine as master machine and other two as nodes

▼ Summary

Number of instances [Info](#)

3

When launching more than 1 instance, [consider EC2 Auto Scaling](#)

Software Image (AMI)

Canonical, Ubuntu, 22.04 LTS, ...[read more](#)
ami-080e1f13689e07408

Virtual server type (instance type)

t2.micro

Firewall (security group)

New security group

Cancel **Launch instance**

After Machines are launched We can rename them as :

Instances (1/3) Info

⌂

Connect

Instance state ▼

Actions ▼

Launch instances

Find Instance by attribute or tag (case-sensitive)

Any state ▼

< 1 >

	Name	Instance ID	Instance state	Instance type	Status check
☐	Ansible-server	i-012d4e05b1366781d	✓ Running	t2.micro	⌂ Initializing
☐	node1	i-0aa1a2fa83f7d2b0d	✓ Running	t2.micro	⌂ Initializing
☒	node2	i-0223294fb8fb0252e	✓ Running	t2.micro	⌂ Initializing

Step 2: Now we will connect to these machines

We can connect to the machine directly via browser

Select the ansible-server and click on connect

rch

[Alt+S]

N. Virginia ▼

Corestack_Role/akshu2079

Instances (1/3) Info

⌂

Connect

Instance state ▼

Actions ▼

Launch

Find Instance by attribute or tag (case-sensitive)

Any state ▼

	Name	Instance ID	Instance state	Instance type	Status c
☒	Ansible-server	i-012d4e05b1366781d	✓ Running	t2.micro	⌂ Initia
☐	node1	i-0aa1a2fa83f7d2b0d	✓ Running	t2.micro	⌂ Initia
☐	node2	i-0223294fb8fb0252e	✓ Running	t2.micro	⌂ Initia

Click on Ec2 instance connect tab

Instance ID
i-012d4e05b1366781d (Ansible-server)

Connection Type

☒ Connect using EC2 Instance Connect
Connect using the EC2 Instance Connect browser-based client, with a public IPv4 address.

☐ Connect using EC2 Instance Connect Endpoint
Connect using the EC2 Instance Connect browser-based client, with a private IPv4 address and a VPC endpoint.

Public IP address
34.201.154.142

Username
Enter the username defined in the AMI used to launch the instance. If you didn't define a custom username, use the default username, ubuntu.

Note: In most cases, the default username, ubuntu, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

Cancel **Connect**

Note: Do not change the username

```
aws Services Search [Alt+S]
```

```
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-36-52:~$
```

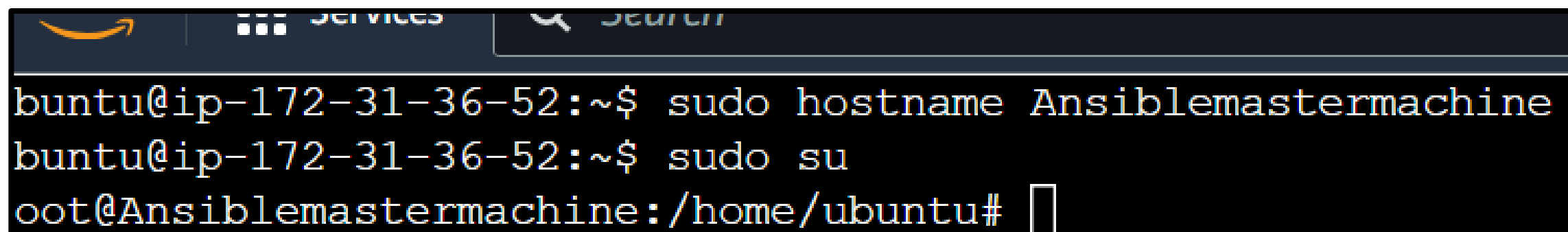
i-012d4e05b1366781d (Ansible-server)

PublicIPs: 34.201.154.142 PrivateIPs: 172.31.36.52

Similarly, We can connect with other machines as well

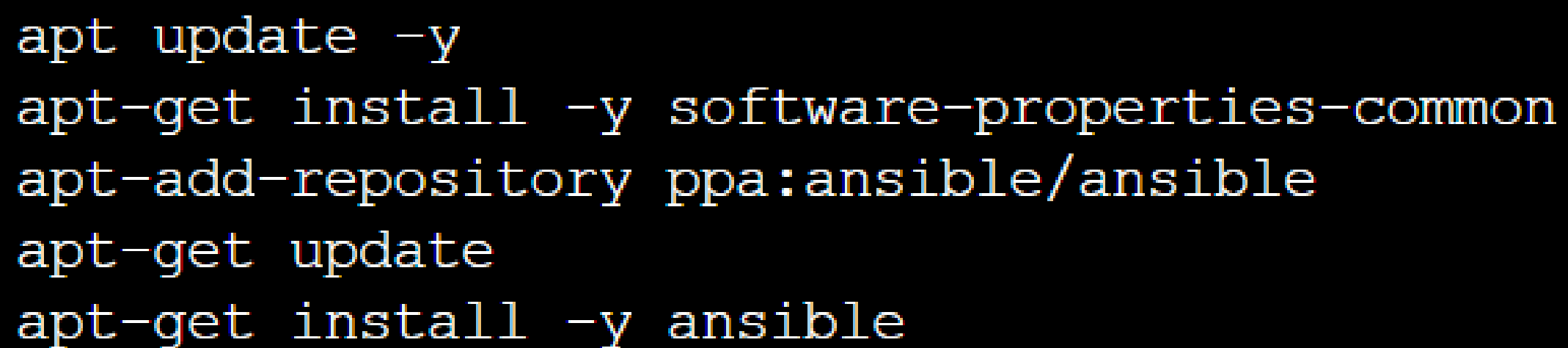
Step 3: lets now install ansible in ansible server (execute the below command only in master machine)

```
sudo su
apt update -y
apt-get install -y software-properties-common
apt-add-repository ppa:ansible/ansible
apt-get update
apt-get install -y ansible
```

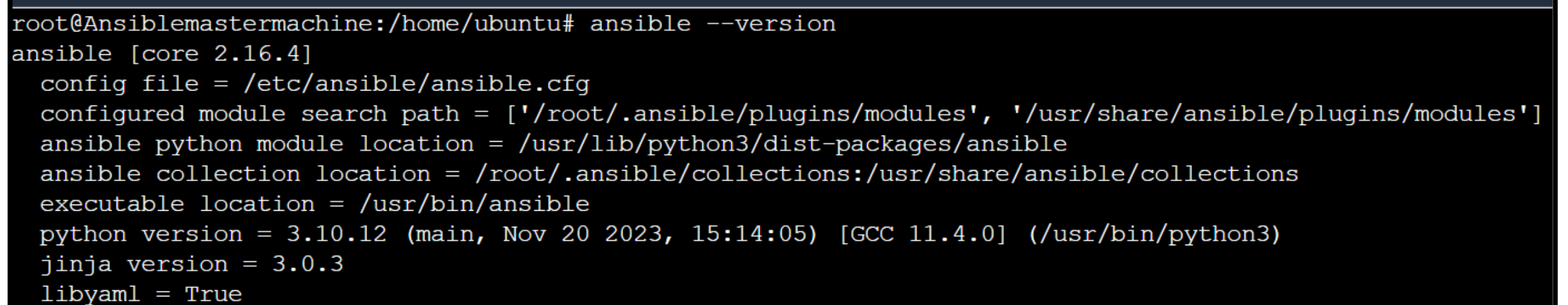


```
buntu@ip-172-31-36-52:~$ sudo hostname Ansiblemastermachine
buntu@ip-172-31-36-52:~$ sudo su
oot@Ansiblemastermachine:/home/ubuntu#
```

ansible --version



```
apt update -y
apt-get install -y software-properties-common
apt-add-repository ppa:ansible/ansible
apt-get update
apt-get install -y ansible
```



```
root@Ansiblemastermachine:/home/ubuntu# ansible --version
ansible [core 2.16.4]
  config file = /etc/ansible/ansible.cfg
  configured module search path = ['/root/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  ansible collection location = /root/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.10.12 (main, Nov 20 2023, 15:14:05) [GCC 11.4.0] (/usr/bin/python3)
  jinja version = 3.0.3
  libyaml = True
```

Step 4: we will now define the hosts from the master machine (ansible server)

vi /etc/ansible/hosts

[ansiblegroup]
Privateip of node1
Privateip of node2

```
vi /etc/ansible/hosts
└
```

Copy the private ip of the node 1

	Name	Instance ID	Instance state	Instance type	Sta
☐	node2	i-0223294fb8fb0252e	Running	t2.micro	
☐	Ansible-server	i-012d4e05b1366781d	Running	t2.micro	
☒	node1	i-0aa1a2fa83f7d2b0d	Running	t2.micro	

Instance: i-0aa1a2fa83f7d2b0d (node1)

Details

Status and alarms [New](#)

Monitoring

Security

Networking

Storage

Tags

▼ Instance summary [Info](#)

Instance ID

i-0aa1a2fa83f7d2b0d (node1)

Public IPv4 address

18.212.151.208 | [open address](#)

Private IPv4 addresses

172.31.42.217

Private IPv4 address copied


```
# This is the default ansible 'hosts' file.
#
# It should live in /etc/ansible/hosts
#
# - Comments begin with the '#' character
# - Blank lines are ignored
# - Groups of hosts are delimited by [header] elements
# - You can enter hostnames or ip addresses
# - A hostname/ip can be a member of multiple groups
[ansibledemo]
172.31.42.217
172.31.39.236

# Ex 1: Ungrouped hosts, specify before any group headers:

## green.example.com
## blue.example.com
## 192.168.100.1
## 192.168.100.10
-- INSERT --
i-012d4e05b1366781d (Ansible-server)
```

(we have to also copy the paste the private ip of the node2 as well)

Step 5: Create a user in Ansible server (master machine) and the nodes

```
# adduser devops
(put password as devops)
And press enter three times and press y
```

```

root@Ansiblemastermachine:/home/ubuntu# adduser devops
Adding user `devops' ...
Adding new group `devops' (1001) ...
Adding new user `devops' (1001) with group `devops' ...
Creating home directory `/home/devops' ...
Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for devops
Enter the new value, or press ENTER for the default
    Full Name []:
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] y

```

Similary create the same username and pass In the nodes as well

Perform same task in node2 as well

(use same username and pass in master and the nodes)

Step 6: We will now configure sshd configuration in master and node machines

#vi/etc/ssh/sshd_config

press i

```

root@Ansiblemastermachine:/home/ubuntu# vi /etc/ssh/sshd_config

```

On line 34 change to PermitRootlogin yes and remove #

```

#ListenAddress 0.0.0.0

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
PermitRootLogin yes
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication yes
-- INSERT --

```

Remove # from line 47

```
#MaxAuthTries 6
#MaxSessions 10

PubkeyAuthentication yes

# Expect .ssh/authorized_keys2 to be disregarded by default in future.
#AuthorizedKeysFile .ssh/authorized_keys .ssh/authorized_keys2

#AuthorizedPrincipalsFile none
-- INSERT --
```

On line 66 enable PasswordAuthentication as yes by removing #

```
# To disable tunneled clear text passwords, change to no here!
PasswordAuthentication yes
#PermitEmptyPasswords no

# Change to yes to enable challenge-response passwords (beware issues with
# some PAM modules and threads)
KbdInteractiveAuthentication no

# Kerberos options
#KerberosAuthentication no
#KerberosOrLocalPasswd yes
#KerberosTicketCleanup yes
#KerberosGetAFSToken no

-- INSERT --
```

(perform same activity in node1 and node2 as well)

(with latest ubuntu these steps need to be added up in master and nodes)

#vi /etc/ssh/sshd_config.d/60-cloudimg-settings.conf

Change password authentication to yes

```

PasswordAuthentication yes
~
~
```

#service ssh restart

```
IS THE INFORMATION CORRECT? [Y/n] y
root@Ansiblemastermachine:/home/ubuntu# vi /etc/ssh/sshd_config
root@Ansiblemastermachine:/home/ubuntu# service sshd restart
root@Ansiblemastermachine:/home/ubuntu#
```

Step 6: We will now give sudo permission to the “devops” user in ansible master and the nodes

#visudo


```
@Ansiblemastermachine:/home/ubuntu# visudo
```

Scroll down to user privilege specification and add below line

```
devops ALL=(ALL:ALL) NOPASSWD:ALL
```

```
# Cmnd alias specification

# User privilege specification
root    ALL=(ALL:ALL) ALL
devops  ALL=(ALL:ALL) NOPASSWD: ALL
# Members of the admin group may gain root privileges
%admin  ALL=(ALL) ALL

# Allow members of group sudo to execute any command
%sudo   ALL=(ALL:ALL) ALL

# See sudoers(5) for more information on "#include" directives:
```

Press ctrl x and then press Y and enter

(We need to perform the same activity for node1 and node2)

Step 7: From Ansible server we will generate the trust relationship (we will generate a key in the master and paste it in the ansible nodes to establish the ssh connection w/o putting password) with the nodes

In Ansible-server (master machine):

```
# su - devops
```

```
# ssh-keygen
```

(above command generate the key in master machine)

(press enter three times)

```

root@Ansiblemastermachine:/home/ubuntu# su - devops
devops@Ansiblemastermachine:~$ ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/home/devops/.ssh/id_rsa):
Created directory '/home/devops/.ssh'.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/devops/.ssh/id_rsa
Your public key has been saved in /home/devops/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:5FN4V674s15pMuk7g5cuzhEgEiIFba7B6okOi08nUIU devops@Ansiblemastermachine
The key's randomart image is:
+---[RSA 3072]-----+
|o+o.o          . |
| .Eo .      . o |
|.o. . . + o . . |
|_+_.+_+_.+_+_.+_+_|

```

Now we need to copy the keypair in the node 1 and node2

Go to Ansible-server(master)

```
# ls -a
```

```
# cd .ssh
```

```

akshat@Ansiblemastermachine:~$ ls -a
.  ..  .bash_logout  .bashrc  .profile  .ssh
akshat@Ansiblemastermachine:~$ cd .ssh

```

```
# ssh-copy-id devops@privateipofnode1
```

```

devops@Ansiblemastermachine:~/.ssh$ ssh-copy-id devops@172.31.24.137
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/devops/.ssh/id_rsa.pub"
The authenticity of host '172.31.24.137 (172.31.24.137)' can't be established.
ECDSA key fingerprint is SHA256:wctGpvtTfVc8XK6WQn05HaLSCg1SmqrZn12CSbmYrbw.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
devops@172.31.24.137's password:

Number of key(s) added: 1

Now try logging into the machine, with:  "ssh 'devops@172.31.24.137'"
and check to make sure that only the key(s) you wanted were added.

devops@Ansiblemastermachine:~/.ssh$ █

```

In the password put the password which we set while creating the user devops

Similarly copy to the node2 as well

```
devops@Ansiblemastermachine:~/.ssh$ ssh-copy-id devops@172.31.23.147
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/devops/.ssh/id_rsa.pub"
The authenticity of host '172.31.23.147 (172.31.23.147)' can't be established.
ECDSA key fingerprint is SHA256:Ai4nqA+kvRqnwYIk6my8nV6ELL9aLG4sywmWTeRMpw.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
devops@172.31.23.147's password:

Number of key(s) added: 1

Now try logging into the machine, with:  "ssh 'devops@172.31.23.147'"
and check to make sure that only the key(s) you wanted were added.

devops@Ansiblemastermachine:~/.ssh$ █
```

We have now established the trust relationship of the master with the nodes.

Step 8: Lets now check if we are able to see the create files in the nodes via ansible

```
devops@Ansiblemastermachine:~$ ansible all -a"touch file1"
172.31.24.137 | CHANGED | rc=0 >>

172.31.23.147 | CHANGED | rc=0 >>

devops@Ansiblemastermachine:~$ ansible all -a"ls"
172.31.23.147 | CHANGED | rc=0 >>
file1
172.31.24.137 | CHANGED | rc=0 >>
file1
devops@Ansiblemastermachine:~$ █
```

